MAIS PAC

IPhO 02.09: Oscillations and Resonance · Lecture + Problems

Oscillations and Resonance

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

1. What an oscillation is
2. Energy and the amplitude
3. The pendulum and other "not-quite-springs"
4. Damping
5. Driven oscillations and resonance
6. Problems: reason, calculate, check
7. Further practice and provenance
8. Where to learn more

1. What an oscillation is

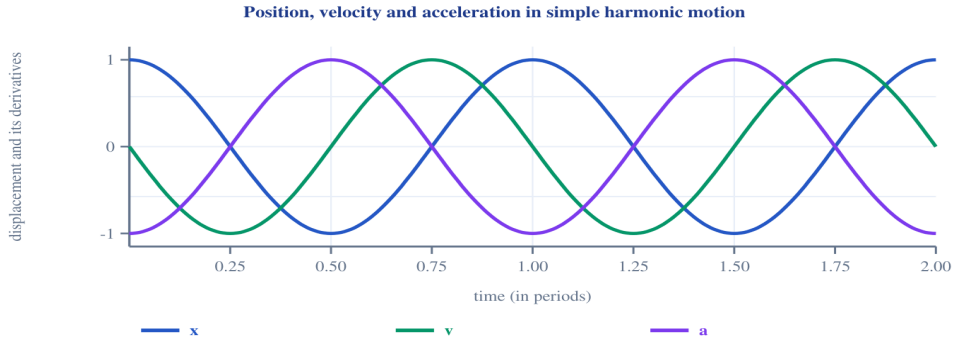
An oscillation happens whenever a system is pushed away from a stable equilibrium and something pulls it back, with the pull growing as the displacement grows. The simplest and by far the most important model is

$$F = -kx \quad m(d^2x)/(dt^2) = -kx \quad (d^2x)/(dt^2) = -\omega^2 x, \quad \omega = \sqrt{((k)/(m))}. \tag{1}$$

This is **simple harmonic motion** (SHM), and its solution is

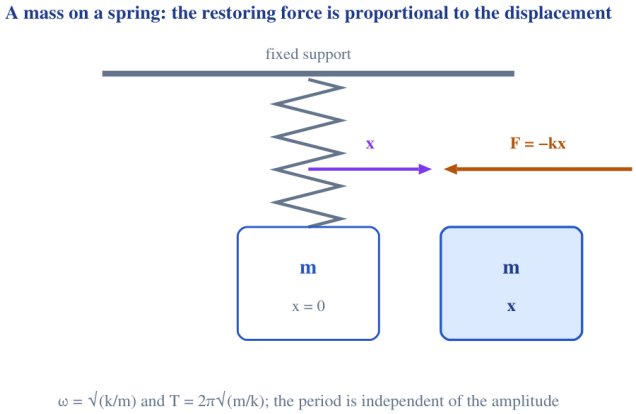
$$x = A\cos(\omega t + \varphi), \quad v = -A\omega\sin(\omega t + \varphi), \quad a = -A\omega^2\cos(\omega t + \varphi) = -\omega^2 x. \tag{2}$$

Two features make SHM special: the period $T = 2\pi/\omega = 2\pi\sqrt{(m/k)}$ does **not** depend on the amplitude, and the motion is the projection of uniform circular motion — which is why phasors, sines and cosines all appear in the same problems.



Velocity leads position by a quarter of a cycle and acceleration is exactly opposite to position.

Figure. Velocity leads position by a quarter of a cycle and acceleration is exactly opposite to position.



ω = √(k/m) and T = 2π√(m/k); the period is independent of the amplitude

Figure 2. The larger the displacement, the larger the restoring force, so the acceleration grows in proportion and the motion repeats in exactly the same time. That is what makes the oscillation isochronous.

The general recipe for any mechanical system near a stable equilibrium is a small-displacement expansion. Write the potential energy, expand $U(x) \approx U(0) + \frac{1}{2}U''(0)x^2$, and read off the effective stiffness $k_{\text{eff}} = U''(0)$. For a mass on a spring, a pendulum, a ball in a bowl or a molecule in a bond, this gives the same equation with different constants.

2. Energy and the amplitude

For a spring, the potential energy is $\frac{1}{2}kx^2$ and the kinetic energy is $\frac{1}{2}mv^2$; their sum

$$E = \frac{1}{2}kA^2 = \frac{1}{2}mv_{\max}^2 \tag{3}$$

is constant. That gives the useful relations $v = \omega\sqrt{A^2 - x^2}$ and $v_{\max} = \omega A$.

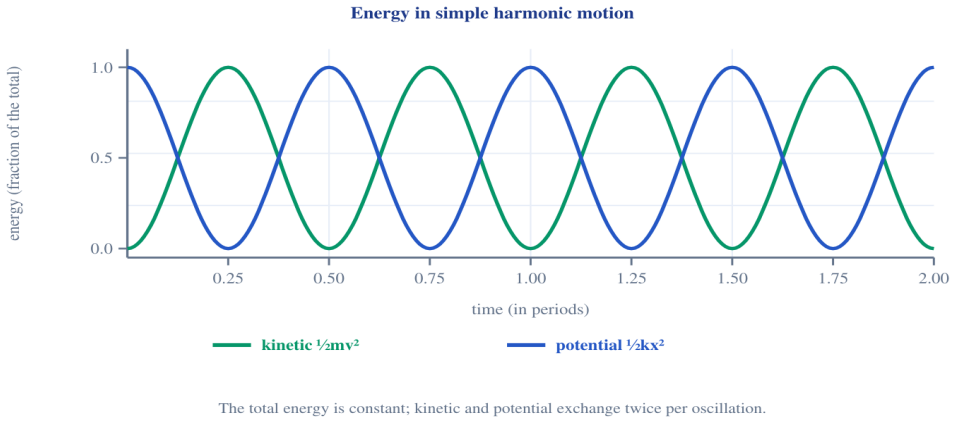


Figure. The total energy is constant; kinetic and potential exchange twice per oscillation.

Because the restoring force is conservative, an oscillation about equilibrium lasts for ever in the ideal case. Anything that dissipates energy — friction, air resistance, electrical resistance — damps it.

3. The pendulum and other "not-quite-springs"

For a simple pendulum of length L, the tangential force is -mgsin θ, so

$$mL\theta = -mg\sin\theta \; \theta = -(g)/(L)\sin\theta. \tag{4}$$

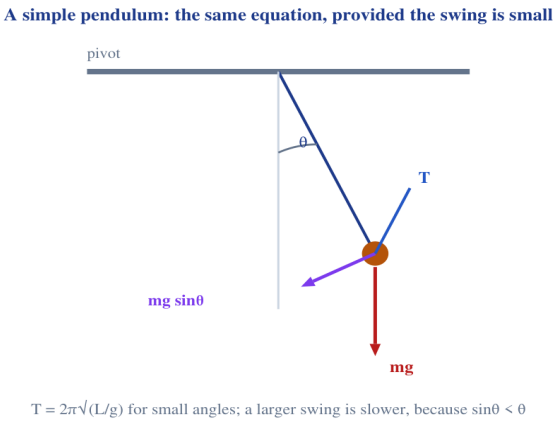


Figure 3. Only the tangential component of the weight drives the motion, and it is mg sinθ. Dividing by the mass gives a = –g sinθ ≈ –(g/L)s, which is the spring equation with k/m = g/L.

Only for small angles, where sin θ≈θ, does this become SHM with ω = √(g/L) and T = 2π√(L/g) — and notably **no dependence on the mass**. For large swings the period grows: the next term in the expansion gives

$$T \approx T_0(1 + (\theta_0^2)/(16)), \tag{5}$$

so a 60° swing is about 7% slower than the small-angle formula predicts. That is why pendulum clocks are driven at small amplitude.

A **physical pendulum** (any rigid body pivoted about a point a distance d from its centre of mass) obeys

$$\omega = \sqrt{((mgd)/(I))}, \tag{6}$$

where I is the moment of inertia about the pivot. A uniform rod of length L pivoted at one end has $I = (1)/(3)mL^2$ and d = L/2, so $T = 2\pi\sqrt{(2L/(3g))}$ — the same as a simple pendulum of length (2)/(3)L.

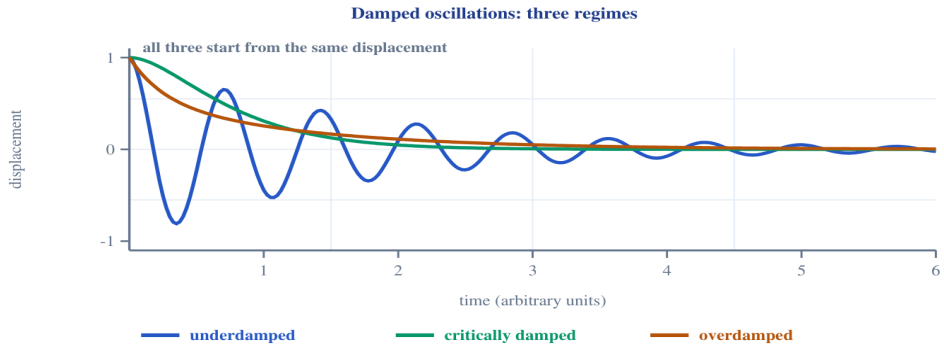
4. Damping

Adding a velocity-dependent drag -b x changes the equation to

$$x + 2\gamma x + \omega_0^2 x = 0, \gamma = (b)/(2m), \tag{7}$$

with three outcomes:

- **Underdamped** ($\gamma < \omega_0$): $x = Ae^{(-\gamma t)}\cos(\omega_d t + \varphi)$ with $\omega_d = \sqrt{(\omega_0^2 - \gamma^2)}$. The amplitude decays exponentially with time constant $1/\gamma$, while the frequency drops slightly.
- **Critically damped** ($\gamma = \omega_0$): the fastest possible return to equilibrium with no overshoot. This is what a well-designed door closer or galvanometer does.
- **Overdamped** ($\gamma > \omega_0$): a slow, creeping return with no oscillation.



Weak damping lets the swing decay slowly; critical damping returns to rest fastest without overshooting.

Figure. Weak damping lets the swing decay slowly; critical damping returns to rest fastest without overshooting.

The **quality factor** $Q = \omega_0/(2\gamma)$ measures how many oscillations survive: roughly Q/π of them before the amplitude falls by e. Energy decays twice as fast as amplitude, $E \propto e^{(-2\gamma t)}$.

5. Driven oscillations and resonance

Now push the oscillator with a sinusoid of frequency ω:

$$x + 2\gamma x + \omega_0^2 x = (F_0)/(m)\cos\omega t. \tag{8}$$

After transients die away the response is sinusoidal at the **driving** frequency, with amplitude

$$A(\omega) = (F_0/m)/(\sqrt{((\omega_0^2-\omega^2)^2 + 4\gamma^2\omega^2)}). \tag{9}$$

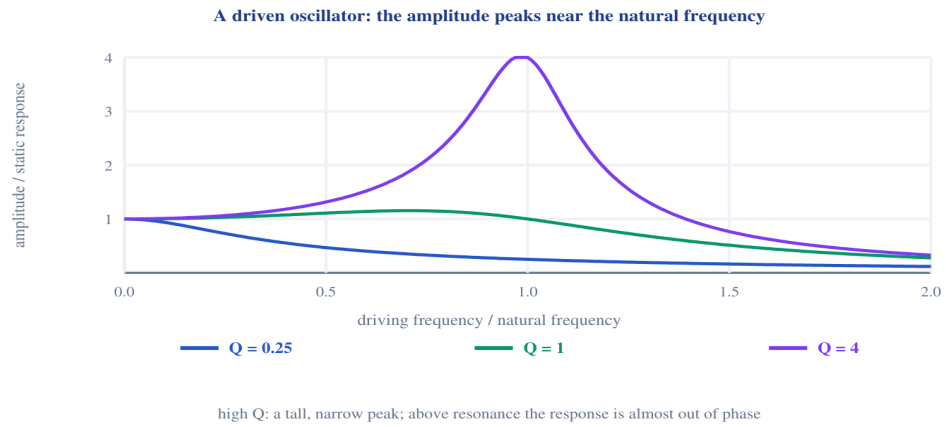


Figure 6. With weak damping the peak sits at ω ≈ ω■ and its height is about Q times the static response; stronger damping lowers the peak and shifts it slightly downwards.

- At low frequency the oscillator follows the force almost in phase, with A≈ F_0/(mω_0²) — the static response.
- At resonance the amplitude peaks, near ω_0 for weak damping, with A_(max)≈ QF_0/(mω_0²). Only damping limits the height.
- At high frequency the inertia dominates, the amplitude falls as 1/ω², and the response is almost exactly out of phase.

Resonance is not a curiosity: it is how a radio selects one station, how a microwave heats water, how a swing is pushed, and how the Tacoma Narrows bridge was destroyed. The same mathematics describes an RLC circuit with L m, 1/C k, R b.

Interactive experiment: A simple pendulum, exactly — open the online handout to use it.

The oscillation checklist. Find the equilibrium, then the restoring force for a small displacement → identify ω from x = -ω²x (never memorise a formula you can derive in two lines) → use energy for amplitudes and speeds, forces for accelerations → for real systems, ask whether damping is weak (exponentially decaying sine) or strong (no oscillation) → for driven systems, compare the driving frequency with ω_0 and remember Q sets the peak height and the width.

6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — A mass on a spring. A 0.50 {kg} mass hangs on a spring of stiffness 80 {N,m^(-1)}. It is pulled 12 {cm} below equilibrium and released. (a) Find the angular frequency, the period and the frequency. (b) Find the maximum speed and the maximum acceleration. (c) Find the speed when the displacement is 6.0 {cm}.

Worked solution →

[1] Problem 2 — A pendulum clock. A clock's pendulum has a period of exactly 2.000 {s} at g = 9.810 {m,s^(-2)}. (a) Find its length. (b) Find the period on the Moon, where g = 1.62 {m,s^(-2)}. (c) Find the fractional change in length needed to correct a clock that loses 10 {s} per day in a place where g is slightly smaller.

Worked solution →

[2] Problem 3 — A physical pendulum. A uniform rod of length 0.80 {m} and mass 0.60 {kg} is pivoted about one end and swings with small amplitude. (a) Find the moment of inertia about the pivot and the period. (b) Find the length of the simple pendulum with the same period. (c) Explain what changes if the rod is instead pivoted about its centre.

Worked solution →

[2] Problem 4 — Energy in SHM. A 2.0 {kg} mass oscillates on a spring of stiffness 200 {N,m^(-1)} with amplitude 15 {cm}. (a) Find the total energy. (b) Find the kinetic and potential energies at x = 5.0 {cm}. (c) Find the positions at which the kinetic energy is three times the potential energy.

Worked solution →

[2] Problem 5 — Damping. An oscillator has m = 0.20 {kg}, k = 32 {N,m^(-1)} and a damping constant b = 0.32 {kg,s^(-1)}. (a) Find ω_0, γ and the quality factor. (b) Find the damped frequency and the fractional change from ω_0. (c) Find the time for the amplitude to fall to one tenth, and the fraction of energy remaining then.

Worked solution →

[2] Problem 6 — A driven oscillator. A 0.10 {kg} mass on a k = 40 {N,m^(-1)} spring is driven by a force of amplitude 0.50 {N} at variable frequency, with damping γ = 1.0 {s^(-1)}. (a) Find the natural frequency. (b) Find the amplitude at $\omega = \omega_0$ and at $\omega = 3\omega_0$. (c) Explain why the amplitude at high frequency falls as $1/\omega^2$ even though the force is applied just as strongly.

Worked solution →

[2] Problem 7 — A buoy in water. A cylindrical buoy of cross-sectional area A = 0.20 {m²} and mass M = 60 {kg} floats in water. It is pushed down slightly and released. (a) Show that the motion is simple harmonic and find its period. (b) Find the period if the buoy is pushed down and released with a large amplitude — qualitatively, does it get longer or shorter? (c) Explain why a wide, low buoy is very stable but a narrow, tall one can capsize.

Worked solution →

[3] Problem 8 — Two springs. A block of mass m is attached to two identical springs of stiffness k, one on each side, and rests on a frictionless surface. (a) Find the period when the block is displaced along the line of the springs. (b) Find the period if the springs are instead attached side by side to a wall and connected to the block in parallel (both pulling the same way). (c) Explain why T changes by a factor of √(2) when the springs are removed and replaced by a single one of stiffness 2k.

Worked solution →

[3] Problem 9 — The large-angle pendulum. A pendulum of length 1.00 {m} is released from rest at 60°. (a) Find the period from the small-angle formula. (b) Find the period to second order in the amplitude and compare with the exact value 2.15 {s}. (c) Find the speed of the bob as it passes through the lowest point, both by energy conservation and by the SHM formula v_(max) = ω A, and explain the difference.

Worked solution →

[3] Problem 10 — Resonance in a real system. A car of mass 1200 {kg} rests on four identical suspension springs, each of stiffness 2.5×10^4 {N,m^(-1)}, with shock absorbers giving a damping constant b = 1.2×10³ {kg,s^(-1)}. (a) Find the natural frequency of the body on its springs. (b) Find the quality factor and decide whether the suspension is close to critical damping. (c) At what speed would a road with regularly spaced ridges 4.0 {m} apart give the worst ride, and what would reduce the resonance?

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

MIT OpenCourseWare 8.03, Vibrations and Waves

Oscillators, resonance, superposition and optics with demonstrations.